INCH-POUND
MIL-DTL-83429C
8 February 2022
SUPERSEDING
MIL-C-83429B
26 September 1990

MILITARY SPECIFICATION

CLOTH, PLAIN AND BASKET WEAVE, ARAMID

This specification is approved for use by all Department and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the requirements for an aramid blend, plain and basket weave, flame resistant cloth that is either solid colored or camouflage printed.

1.2 Classification. The cloth covers the following types and classes as specified. (see 6.2).

1.2.1 Types.

Type I – 2 X 2 Basket weave, solid color, aramid blend
Type II – Plain weave, aramid blend

1.2.2 Classes.

Class 1 – DELETED
Class 2 – DELETED
Class 3 – DELETED
Class 4 – DELETED
Class 5 – DELETED
Class 6 - Solid color
Class 7 – DELETED
Class 8 – DELETED
Class 9 – Operational camouflage pattern (OCP)

Comments, suggestions, or questions on this document should be addressed: Attn: DLA Troop Support, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil.

AMSC N/A FSC 8305

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTION

A-A-55195 - Thread, Para-aramid, Spun, Intermediate Modulus

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in these documents are cited in the solicitation or contract.

DRAWINGS

U.S. ARMY DEVCOM, SOLDIER CENTER

2-1-2592 - Operational Camouflage Pattern (OCP)

(Copies of drawings are available from the U.S. Army DEVCOM, Soldier Center, ATTN: FCDD-SCD-PEI, 10 General Greene Avenue, Natick, MA 01760-5019.)

ENVIRONMENTAL PROTECTION AGENCY

EPA Pollutants/toxics subtopics

(Copies are available online at https://www.epa.gov/regulatory-information-topic.)

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CODE OF FEDERAL REGULATIONS (CFR)

Title 16 CFR Part 303 Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1 - Evaluation Procedure for Gray Scale for Color Change
AATCC EP8 - Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM8 - Test Method for Colorfastness to Crocking: Crockmeter
AATCC TM15 - Test Method for Colorfastness to Perspiration
AATCC TM16.3 - Test Method for Colorfastness to Light: Xenon-Arc
AATCC TM20A - Test Method for Fiber Analysis: Quantitative
AATCC TM61 - Test Method for Colorfastness to Laundering: Accelerated
AATCC TM81 - Test Method for pH of the Water-Extract from Wet Processed Textiles
AATCC TM135 - Test Method for Dimensional Changes of Fabrics After after Home Laundering

(Copies of these documents are available online at https://aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D737 - Standard Test Method for Air Permeability of Textile Fabrics
ASTM D1423 - Standard Test Method for Twist in Yarns by Direct-Counting
ASTM D1683/D1683M - Standard Test Method for Failure in Sewn Seams of Woven Fabrics
ASTM D1907/D1907M - Standard Test Method for Linear Density of Yarn (Yarn Number) by the Skein Method
ASTM F1930 - Standard Test Method for Evaluation of Flame Resistant Clothing for Protection Against Flash Fire Simulations Using an Instrumented Manikin

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ASTM D3775	- Standard Test Method for Warp (End) and Filling (Pick) Fabric Count of Woven Fabric
ASTM D3776/D3776M	- Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D5034	- Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D5103	- Standard Test Method for Length and Length Distribution of Manufactured Staple Fibers (Single-Fiber Test)
ASTM D6413/D6413M	- Standard Test Method for Flame Resistance of Textiles (Vertical Test)

(Copies of these documents are available online at https://www.astm.org.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 2112 - Standard on Flame-Resistant Garments for Protection of Industrial Personnel Against Flash Fire, 2007 edition

(Copies of this document are available online at https://www.nfpa.org.)

OTHER PUBLICATIONS

Fabric Defect Replica Scales

(For Information on obtaining copies of the Fabric Defect Replica Scales please contact U.S. Army DEVCOM, Soldier Center, ATTN: FCDD-SCP-PEI, 10 General Greene Avenue, Natick, MA 01760-5019.)

INFORMA HEALTHCARE

Repeat Insult Patch Test - Modified Draize Procedure –
Principles and Methods of Toxicology, A Wallace Hayes (editor).

(Copies of this document are available online at https://www.crcpress.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2) a sample shall be subjected to first article inspection in accordance with 4.2.

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3.2 Standard sample. The finished cloth shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Material.

3.4.1 Aramid fiber blend. The cloth blend for both Types and all Classes shall consist of 92 percent meta-aramid fiber, 5 percent para-aramid fiber, and 3 percent conductive fiber (see 6.4). The aramid fibers shall have a nominal denier of 1.5 to 1.7 denier per filament, a staple length of 1-1/2 to 2-inches, and shall not char at a temperature less than 675°F. The conductive fiber shall be nylon sheathed carbon core fiber and shall be 3 denier per filament, and 1-1/2 to 2-inches in staple length. Testing shall be as specified in 4.5.

3.4.2 Yarn. The yarn for both the warp and filling shall be spun into singles for the Type I cloth, and 2-ply for the Type II, cloth. Testing shall be as specified in 4.5 (see 6.4.1).

3.4.3 Weave. The Type I cloth shall be a 2 X 2 basket weave and the Type II cloth shall be plain weave.

3.5 Color.

3.5.1 Type I and Type II, Class 6, solid colors. The color of the finished cloth as specified in the contract or procurement documents (see 6.2), shall be solid Camouflage Green 483, Sage Green 1659, Tan 1654, Tan 380, Foliage Green 504, Tan 499, or natural, and shall match the standard sample.

3.5.2 Type II, Class 5, Woodland camouflage pattern print. DELETED

3.5.3 Type II, Class 7, Desert camouflage pattern print (3-Color). DELETED

3.5.4 Type II, Class 9, Operational Camouflage Pattern (OCP). The finished cloth shall be either dyed to a ground shade matching Dark Cream 559 and then overprinted with the remaining six (6) colors Tan 525, Light Sage 560, Olive 527, Dark Green 528, Brown 529, and Bark Brown 561 areas of the camouflage pattern (see Drawing 2-1-2592) or when the ground shade is not dyed to Dark Cream 559 all seven (7) colors of the camouflage pattern shall be printed to match all seven (7) colors of the pattern. Resin bonded pigments shall not be used.

3.5.5 Visual shade matching. The color and appearance of the dyed or camouflage printed finished cloths shall match the standard sample when tested as specified in 4.5.

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3.5.6 Colorfastness (all Classes). The finished cloth shall conform to the colorfastness requirements specified in Table I when tested as specified in 4.5.

TABLE I. Colorfastness requirements (Classes 6 and 9).

Colors Evaluation	Laundering Color Change (3 cycles) (min.)	Light (after 40 AFU or 170 kJ/(m^2nm) @ 420 nm) 1/ (min.)	Perspiration (acid & alkaline) Color Change (min.)	Crocking (Dry/Wet) (min.)
All Colors: Type I & Type II, Classes 6 and 9 except Natural	3	3	4	3.5

1/ AFU – AATCC Fading Units

3.6 Pattern execution. The pattern on the printed finished cloth(s) shall match the standard sample with respect to design, colors, and registration of the respective areas (see 6.3). The various areas of the pattern shall be properly registered in relation to each other and shall present definite sharp demarcations and/or feathering as required for each color. Each pattern area shall show full coverage; skitteriness, feathering, haloing or trapping and off register exceeding that shown on the standard sample in any of the printed areas will not be acceptable. When a standard sample is not available for pattern execution, a pattern drawing will be provided and the pattern on the finished cloth shall match that of the specified drawing (see 2.2.2) (see 6.2 and 6.3). The pattern repeat shall be as follows:

Class 9 Operational Camouflage Pattern (OCP) - 25.255 (+1.25, -2.50) inches in the warp direction and the maximum width of the camouflage printed area shall be 68 (± 1.50) inches in the filling direction.

3.7 Spectral reflectance. The spectral reflectance values for each class shall conform to the requirements specified in their respective applicable Tables II and V when tested as specified in 4.5.

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TABLE II. Spectral reflectance values (percent), Class 6.

Wavelength, Nanometers (nm)	Solid Shades							
	Camouflage Green 483		Sage Green 1659		Tan 380 and Tan 1654		Tan 499	
	Min	Max	Min	Max	Min	Max	Min	Max
600	3	10	6	12	N/A	N/A	8	26
620	3	10	6	12	N/A	N/A	8	26
640	3	10	7	15	N/A	N/A	8	30
660	3	12	7	18	N/A	N/A	8	34
680	3	14	8	18	N/A	N/A	12	38
700	4	28	8	18	25	53	12	40
720	5	40	12	22	25	54	16	46
740	10	52	16	28	25	55	22	50
760	18	56	22	35	26	56	30	50
780	20	56	28	40	27	57	34	54
800	24	58	32	45	28	58	36	56
820	30	60	36	52	30	59	38	58
840	38	60	40	58	33	62	38	58
860	40	60	46	64	36	65	40	60

TABLE III. Spectral reflectance values (percent), Class 5. DELETED

TABLE IV. Spectral reflectance values (percent), Class 7. DELETED

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TABLE V. Spectral reflectance (percent), Class 9.

Operational Camouflage Pattern (OCP)						
Wavelength, Nanometers(nm)	Dark Cream 559 and Tan 525		Light Sage 560, Olive 527 and Brown 529		Dark Green 528, and Bark Brown 561	
	Min	Max	Min	Max	Min	Max
600	22	44	12	30	3	11
620	24	45	12	30	3	12
640	24	45	12	32	4	13
660	25	45	12	32	4	14
680	28	45	14	34	4	17
700	28	48	14	36	6	23
720	30	52	16	39	6	23
740	32	55	18	41	10	25
760	36	56	20	43	14	30
780	38	57	22	45	18	35
800	40	57	22	45	21	40
820	44	58	24	46	24	42
840	46	59	26	47	26	43
860	48	60	28	48	28	45

3.8 Physical Requirements. The physical requirements of the finished cloth shall be as specified in Table VI when tested as specified in 4.5.

TABLE VI. Physical requirements.

Characteristic	Type I	Type II	
		Class 6	Classes 9
Weight, oz./sq. yd.	5.0 (min.)	4.3 - 5.0	4.7 – 6.0
Yarns per inch (min.):			
Warp	90	70	66
Filling	88	47	44
Breaking strength, lbs. (min.):			
Warp	100	180	180
Filling	100	100	100
Tearing strength, lbs. (min.):			
Warp	15	12	8
Filling	15	8	5

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TABLE VI. Physical requirements Continued.

Characteristic	Type I	Type II	
		Class 6	Classes 9
Flame resistance, 1/			
Initial and after 25, 50, 100 laundering cycles (Class 9):			
After flame, seconds (max.)	2.0	2.0	2.0
After glow, seconds (max.)	25.0	25.0	25.0
Char length, inches (max.)	4.5	4.5	4.5
Initial and after 25 launderings (Sage Green 1659 and Tan 1654 Class 6)			
After flame, seconds (max.)	2.0	2.0	2.0
After glow, seconds (max.)	25.0	25.0	25.0
Char length, inches (max.)	4.5	4.5	4.5
Thermal shrinkage resistance, percent (max.)			
Initial only	10	10	10
Heat Transfer Performance (HTP) (formerly Thermal Protective Performance (TPP))			
Spaced (min.)	10.0	10.0	10.0
Contact (min.)	7.0	7.0	7.0
Flash fire (Predicted Burn Injury), percent (max.)			
With head	35	35	35
Without head	28	28	28

1/ No melt or drip

3.8.1 Width. The width of the finished cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of the selvage when fly-shuttle looms or shuttleless looms with a tuck-in selvage are used. For all other shuttleless looms, the width measurement shall be made between the last warp yarn on each side excluding the protruding fringe(s).

3.8.2 Fabric break open. The flame from Meker burner shall not pass from the flame contact side to the other side of the fabric due to the fabric breaking open when tested as specified in 4.5.

3.9 Electrostatic decay. The finished cloth shall, when charged toward +5000 volts and -5000 volts, reach a minimum level of +4000 volts and -4000 volts respectively and shall dissipate 90 percent of this charge in no more than 1/2 second. The average warp results and average filling

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results of the face and back shall meet these requirements before and after five (5) launderings when tested as specified in 4.6.3.

3.10 pH. The pH value of the water extract shall be not less than 5.0 nor more than 8.5 when tested as specified in 4.5.

3.11 Curling. The finished cloth shall lie flat, without distortion, and shall show no evidence of curling when tested as specified in 4.6.4.

3.12 Dimensional change. The Type I and Type II, Class 6 cloth shall not shrink more than 4.0 percent in the warp direction nor more than 1.5 percent in the filling direction, after fifteen launderings. The shrinkage in the warp or filling directions for Type II, Class 9 cloth shall be not greater than 3.5 percent for individual specimen and not greater than 3.0 percent for the sample unit average after five (5) launderings. Testing shall be as specified in 4.5. The pre-shrinking process shall not be identified by name or trademark, either on the cloth, ticket or package.

3.13 Seam efficiency. The finished cloth shall have a seam efficiency of not less than 80 percent when tested as specified in 4.5.

3.14 Length and put-up. For Government procurement only, unless otherwise specified (see 6.2), the finished cloth shall be furnished in continuous lengths, each not less than 40 yards.

3.15 Face identification for Type I, and Type II Class 6 cloths. The face side of these cloths shall be identified by applying a stamping on that side of the cloth with the word “Face” at each end of the roll.

3.16 Fiber identification. Each roll of cloth shall be ticketed for fiber content in accordance with 16 CFR Part 303 Rules and Regulations Under the Textile Fiber Products Identification Act.

3.17 Toxicity. The finished aramid cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.7. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.18 Workmanship. The finished cloth shall conform to the quality of product established by this specification. The occurrence of defects as specified in 4.3 and 4.4, shall not exceed the quality acceptance levels as specified in the contract or procurement document.

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4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified or qualified in this specification or applicable procurement documents.

4.3 Types of inspection.

4.3.1 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance, color and finished defects specified in 4.4, and tested for the characteristics as specified in 4.5.

4.3.2 Conformance inspection. Conformance inspection shall include the examinations of 4.4 and the tests of 4.5 through 4.6 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQL's) as specified in the contract and/ or order, except where otherwise indicated (see 6.2).

4.4 Visual examination. Each roll in the sample shall be examined yard by yard on the face side for defects in accordance with MIL-STD-3064, Type I.

4.4.1 Roll identification and marking examination. During the yard-by-yard examination each roll in the sample shall be examined for roll identification and marking defects as specified in MIL-STD-3064.

4.4.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.4.3 Length examination (individual and total yardage). During the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-3064.

4.5 End item testing. The cloth shall be tested for the applicable characteristics specified in Table VII. The methods of testing specified, wherever applicable, and as listed in Table VII shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be five (5) continuous yards full width of the finished cloth. The lot size shall be expressed in units of 1-yard. The lot shall be unacceptable if one (1) or more sample units fail to meet any test requirement specified.

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TABLE VIII. End item tests.

Characteristic	Requirement Paragraph	Test Method
Material:		
Fiber content	See 3.4.1	AATCC TM20A (see 6.6)
Denier	See 3.4.1	ASTM D1907/D1907M
Staple length	See 3.4.1	ASTM D5103 1/
Non charring	See 3.4.1	1/
Conductive fiber	See 3.4.1	1/
Yarn ply	See 3.4.2	ASTM D1423 2/
Weave	See 3.4.3	Visual 2/
Visual shade matching	See 3.5.5	AATCC EP9, Option C 3/
Colorfastness to: (Type I and II, Class 6 and 9)		
Laundering (Color change only)	Table I	AATCC TM61, Test 3A, 4/, 5/
Light	Table I	AATCC TM16.3 5/
Perspiration (Color change only)	Table I	AATCC TM15 4/, 5/
Crocking	Table I	AATCC TM8, 6/
Pattern execution	See 3.6	Visual 2/
Spectral reflectance	Table II -Table V	See 4.6.1
Weight	Table VI	ASTM D3776/D3776M, Option C
Yarns per inch	Table VI	ASTM D3775
Breaking strength	Table VI	ASTM D5034
Air Permeability	Table VI	ASTM D737
Flame resistance		
Initial	Table VI	ASTM D6413/D6413M
Flame resistance After laundering (25, 50 and 100 cycles) (Classes 6 and 9) After 25 launderings (Sage green 1659, Tan 1654 of Class 6)	Table VI	AATCC TM135, (3)(V) Aiii & ASTM D6413/D6413M 7/, 8/

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TABLE VII. End item tests. – Continued

Characteristic	Requirement Paragraph	Test Method
Thermal shrinkage resistance: Initial	Table VI	NFPA 2112 (section 8.4) 8/
Heat transfer performance (formerly Thermal Protective Performance)	Table VI	NFPA 2112 (section 8.2) 8/
Flash fire (Predicted Burn Injury)	Table VI	ASTM F1930 (4 second exposure) 8/, 9/
Fabric break open	See 3.8.2	See 4.6.2
Electrostatic decay: Initial After five (5) launderings	 See 3.9 See 3.9	 See 4.6.3 AATCC TM135, (3)(V) Aiii, 7/ & 4.6.3
pH	See 3.10	AATCC TM81
Curling	See 3.11	See 4.6.4
Dimensional stability: Type I and Type II, Class 6 After 15 launderings Type II, Class 9 After five (5) launderings	 See 3.12 See 3.12	 AATCC TM135, (3)(V) Aiii 7/, 10/ AATCC TM135, (3)(V) Aiii 7/, 10/
Seam efficiency	See 3.13	ASTM D1683/D1683M 11/
Toxicity	See 3.17	4.7

1/ Unless otherwise specified, a copy of the procurements documents referencing the specified requirement shall be acceptable verification of the stated requirement.
2/ One (1) determination shall be made from each sample unit and the results reported as “pass” or “fail”.
3/ The color and appearance of the finished cloth shall match the standard sample when viewed using the AATCC EP 9, Option C (see 6.5), with sources simulating artificial daylight D_{75} illuminant with a color temperature of 7500K(± 200), illumination of 100 -125 foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856K (± 200).
4/ Except that the specimen shall be evaluated for color change only.
5/ Rated using AATCC EP1.
6/ Rated using AATCC EP8.
7/ The last wash cycle shall be performed without the use of detergent. Specimens shall be dried after each laundering.
8/ The flame resistance testing after 50 and 100 launderings shall be required for first article testing only. The thermal shrinkage resistance, heat transfer performance and flash fire testing shall be required for first article only.
9/ Material in applicable type and class shall be tested in accordance with applicable garment end item specification. The coat or shirt shall be tucked into the waist or trouser. Manufacturer shall submit three (3) sets of size Medium Regular garments to Product Manager Soldier Clothing and

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Individual Equipment (PdM-SCIE) for all Classes, except for Class 6 where three (3) specimens will be sent to the Air Force Uniform Office. Tests will be conducted using standard cotton T-shirts and briefs.
10/ The cloth shall not be pressed after tumble drying prior to measurement
11/ The needle shall measure 0.044 (±0.001) inch across the blade at the eye. The thread for all types shall be aramid in accordance with A-A-55195, Type I, Tex Size 59, 3-ply for the needle Tex Size 39, 3-ply for the bobbin. Seam type LSc-2 or LSs-2.

4.6 Methods of inspection.

4.6.1 Spectral reflectance test. Spectral reflectance data shall be obtained from 600 to 860 nanometers (nm) at 20 nm for all solid colors (Class 6) and Operational Camouflage Pattern (Class 9) shall be 700 to 860 nanometers (nm) at 20 nm intervals on a spectrophotometer relative to the polytetrafluoroethylene (PTFE) family of compounds, the preferred white standard on a spectrophotometer relative to the barium sulfate standard, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white, e.g. magnesium oxide or vitrolite tiles. The spectral band width shall be less than 26 nm at 860 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates either CIE Source A or CIE Source Measurements shall be taken on a minimum of two (2) different areas and the data averaged. The measured areas should be at least 6-inches away from the selvage. The specimen shall be measured as a single layer backed with three (3) layers of the same fabric and shade. The specimen shall be viewed at an angle no greater than 10 degrees from normal, with the specular component included. Specimens shall be oriented in different directions during testing. When possible, the specimens tested shall not contain the same warp or filling yarns when presented to the sample port. Photometric accuracy of the spectrophotometer shall be within 1-percent and wavelength accuracy within 2-nanometers. Camouflage materials should be measured with the appropriate aperture size to ensure that only one (1) color is measured at a time. Always try to use the largest aperture possible. Any color having spectral reflectance values falling outside the limits at four (4) or more of the wavelengths specified shall be considered a test failure.

4.6.2 Fabric break open test. The specimen shall be rigidly held in a horizontal position between two (2) metal plates with a 6-inch diameter fabric exposure. One (1) side of the fabric shall be exposed to a Meker burner in the center at a 90 degree angle using natural gas for 30 seconds with a 2-inch distance between fabric and burner top. Proper flame adjustment shall be accomplished by setting the needle valve at the midpoint between the open and closed positions and adjusting the gas by means of flow meter to 2 liters per minute.

4.6.3 Electrostatic decay test. The specimen shall be tested according to the method specified in 4.6.3.1 through 4.6.3.5 initially and after five (5) laundering cycles.

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4.6.3.1 Apparatus. The apparatus shall consist of a humidity test chamber, a meter for measuring relative humidity and temperature, a dehumidifier, an air-ionizer or static eliminator, a faraday cage, a source capable of out putting 5000 volts, and voltage meters to display the applied voltage and the voltage detected by the sensor. The faraday cage shall contain two parallel electrodes, on which specimens may be mounted and charged, and a sensor that will detect voltage on the specimen surface. A staticmaster is needed to eliminate residual charges of 500 volts or less from the samples. A humidifier and a humidity controller may be added to the system to facilitate conditioning of the sample.

4.6.3.2 Test specimen. The test specimen shall be 3-inch by 6-inch rectangular piece of material. The warp or filling yarns shall run parallel to the 6-inch length of the specimen. The direction of test (warp or filling) shall be along the length of the specimen and each specimen shall be labeled accordingly in one corner. Test specimens shall be cut so that no two contain the same set of warp and filling yarns.

4.6.3.3 Number of determinations. Unless otherwise specified in the procurement document, six (6) specimens shall be tested, three (3) in the warp direction and three (3) in the filling direction, each at both sides: face and back.

4.6.3.4 Procedure. The faraday cage and specimens to be tested shall be preconditioned at 10 (± 2) percent relative humidity overnight, and conditioned at 20 (±2) percent relative humidity for a minimum of 24 hours at 75 (± 5)°F. The specimens shall be tested at 20 (± 2) percent relative humidity and 75 (± 5)°F. An air-ionizing blower or a static eliminator must be used in the chamber while conditioning but must be turned off once testing begins.

a. Allow test equipment one-hour warm-up time. Adjust voltage source for a charging voltage of 5000 volts. Mount a 3-inch by 5-inch aluminum calibration plate across the electrodes in the faraday cage such that it is centered in front of the voltage sensor opening. Apply a charge of 5000 volts, and calibrate the voltage meter designated to indicate detected voltage on the sample equal to 5000 volts. Remove the aluminum plate.

b. Mount a specimen tautly across the electrodes, centering it over the sensor opening. The surface of test specimen (back or front) shall face the sensor. The CUT-OFF must be set to 10 percent. With the high voltage off, depress the CHG button to measure initial charge on the test specimen. If it is not reading zero, it may not be antistatic. A staticmaster shall be used on the specimen before the test to remove residual charge of 500 or less.

c. Apply 5000 volts to the electrodes for a period of 20 seconds. To do this, depress the + button on the right (the reading on the CHARGING VOLTAGE meter must read 5000 volts); then depress the CHG button. Wait 20 seconds; record the highest reading on the SAMPLE CHARGE meter. Then depress the TEST button and from the readout under DECAY TIME-SEC, record how long it takes for the charge to decay from the maximum voltage to 10 percent of the maximum level. Depress the ZERO/STBY button after the test. The high voltage (5000 volts) shall be turned off and the specimen immediately grounded.

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d. If the maximum voltage level on the SAMPLE CHARGE meter did not reach at least 4000 volts during the 20-second charging period, the specimen shall be recorded as non-passing and the reason noted.

e. If the DECAY TIME-SEC is less than 0.5 second, the specimen is acceptable; otherwise it is not and the reason for failure shall be noted.

f. After testing the other side of the same specimen, remove it and repeat 4.7.3.2 through 4.7.3.5 for the front and back sides of the other specimens of the test fabric.

g. Recalibrate periodically, at the minimum between each set of six (6) specimens.

4. 6.3.5 Report and evaluation.

a. The test method used and any alterations to the method shall be reported.

b. Test conditions, including relative humidity, room temperature and conditioning time, shall be stated.

c. Equipment names, model numbers and manufacturers shall be listed. Special modifications to equipment shall be described. A description of any in-house built equipment shall be included.

d. Identification of materials tested, including name, composition, weave, printed/dyed, finishes, manufacturer and other pertinent information shall be included.

e. Alteration to materials shall be noted together with the test method used, or description of the process applied; e.g., laundering.

f. The average time to decay to 10 percent of the maximum voltage shall be included for each of the warp and filling directions, as well as the overall average, for each fabric tested. The maximum voltage level reached by the specimen shall be recorded. It shall be indicated whether the fabric is acceptable or non-acceptable, passing or non-passing.

g. Passing the test requires the following: 1) specimen's reaching at least 4000 volts during the 20-second charging period and 2) less than 0.5-second decay time for the charge from maximum voltage level to 10 percent of maximum level.

4.6.4 Curling test. Two (2) specimen of cloth, 1-1/2-inches wide by 6-inches long shall be cut, one (1) having the long dimension parallel to the warp and the other with the long dimension parallel to the warp and the other with the long dimension parallel to the filling. Both specimen shall be placed on a flat surface for at least five (5) minutes and then visually examined for evidence of curling.

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4.7 Toxicity test. When required, (see 6.2), an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the Aramid cloth is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If the toxicity requirement (see 3.17) can be demonstrated with historical use data on the finishing treatments used, toxicity testing may not be required (see 6.2).

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6.1 Intended use. The cloth is intended for use in the manufacture of special flight, shipboard and ground combat clothing.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type and class of cloth required (see 1.2).
c. The specific issue of individual documents reference (see 2.2 and 2.3).
d. When a first article is required (see 3.1).
e. Color required when Type I or Type II, Class 6 is specified (see 3.5.1).
f. Pattern drawing if required (see 3.6).
g. Width of cloth required (see 3.8.1).
h. Length and put-up required if other than specified (see 3.14).
i. When toxicity testing is required (see 3.17).
j. Conformance inspection conditions (see 4.3).
k. Packaging requirements (see 5.1).

6.3 Standard sample. For access to samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposal.

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6.4 Material. The aramid cloth has been successfully manufactured with yarns produced from an intimate fiber blend of 92 percent Nomex®, 5 percent Kevlar® and 3 percent P-140. The fiber blend is available in bale form from Invista®, Wilmington, DE.

6.4.1 Yarn. Cloth woven with 24/1 yarn for both the warp and filling with a minimum of 90 ends per inch and 88 picks per inch have been used to make the Type I cloth. Cloth woven with 38/2 yarn for both the warp and filling with a minimum of 70 ends per inch and 47 picks per inch have been found to meet the basic material requirements for the Type II, Class 6 cloth.

6.5 Visual shade matching. In 2017, Option A of AATCC EP9 was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens.

6.6 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.7 Subject term (key word) listing.

Camouflage
Electrostatic decay
Flame resistant
Kevlar®
Nomex®
Operational Camouflage Pattern (OCP)

6.8 Changes from previous issue. Marginal notations are not used on this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - GL
Navy - NU
Air Force - 11

Preparing activity:
DLA - CT

(Project 8305-2022-001)

Review activities:
Army - MD
Navy - AS

NOTE: The activities listed above were interested in this document as of date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST online database at https://assist.dla.mi.

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